Venance Komu

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Education

Alfred University Alfred, NY

Bachelor of Science in Electrical Engineering, Mathematics - Double Major; Computer Science Minor *Expected May 2029*

- **Relevant Coursework:** Object-Oriented Programming (OOP), Data Structures and Algorithms, Computer Organization & Architecture, Digital Logic, Circuit Theory, Software Development Lifecycle (SDLC) (Agile, Scrum and Waterfall), System Design, Discrete Mathematics, Linear Algebra, Probability & Statistics, Multivariable Calculus, Differential Equations
- **GPA:** 3.95/4.00 | **Awards & Honors:** Dean's List (2025–2026)

Technical Skills

Programming Languages: Python, C, C++, Java, JavaScript, HTML5/CSS3, SQL, PostgreSQL, MongoDB, MATLAB, VHDL
Frameworks & Libraries: React, AngularJS, Bootstrap, FastAPI, PyTorch, YOLOv8 (Ultralytics), OpenCV, Chart.js, NumPy, pandas, Matplotlib
Tools & Operating Systems: MS Office (Excel, Word, PowerPoint, Outlook), Git, GitHub, GitHub Copilot CLI, REST APIs, Visual Studio, Azure, Claude Code, Linux (Ubuntu), Windows 10/11, macOS, SolidWorks, LTspice

Experience

Alfred University, Inamori School of Engineering May 2026 – Jul 2026

Undergraduate Researcher *Alfred, NY*

- Compiled and cleaned data on secondary gallium sources in **Python (pandas, NumPy)** and **Excel**, applying descriptive statistics to summarize findings, compare materials, and assess their potential for gallium recovery.
- Conducted a **Material Flow Analysis (MFA)** modeling gallium flows and availability across secondary sources, assessing recovery potential and possible contributions to the **U.S. gallium supply chain**.
- Analyzed **X-ray fluorescence (XRF)** measurements with **Python** scripts and **Matplotlib** plots to assess gallium concentrations and compare materials as potential recovery sources, versioning analysis code in **Git/GitHub**.

National Housing Corporation Sep 2024 – Jan 2025

Intern *Moshi, Tanzania*

- Restructured and standardized property and tenant datasets in **Excel** and **SQL**, improving data accuracy, record retrieval, and operational efficiency across property-management processes.
- Analyzed property tax, payment, and maintenance records with **Excel** pivot tables and lookups to identify discrepancies and monitor property requirements, presenting findings in **PowerPoint** to support data-driven management decisions.

Technical & Professional Development

CodePath Spring 2026

Technical Interview Prep Student *Remote (San Francisco, CA)*

- Completed intensive training in **data structures and algorithms** and computational problem-solving in **Python**.
- Implemented **Python** algorithms using **recursion**, **sorting**, **searching**, **hash tables**, and linked data structures.

freeCodeCamp Fall 2026

Machine Learning & Scientific Computing with Python *Remote (Alfred, NY)*

- Built **machine learning** models in **Python** for image classification (**computer vision**), SMS spam detection (**NLP**), and healthcare cost prediction (**regression**).
- Developed a **K-Nearest Neighbors (KNN)** book recommendation system and a Rock Paper Scissors game-playing algorithm, alongside **object-oriented** Python programs for scientific computing.

Projects

Critical Materials Circuit Component Detector | *Python, PyTorch, YOLOv8, OpenCV, NumPy/pandas, Linux* Summer 2026

- Trained a **YOLOv8 computer vision** model in **PyTorch** to detect 5 component classes (ICs, capacitors, resistors, transistors, connectors) on about **500 PCB images** with **2,000+ labeled instances**, split 70/15/15.
- Matched detections to a **critical-materials** reference database to estimate each board's material content (**mg or g per PCB**), stating the assumptions behind each estimate.

Collaborative Food Cargo Tracking System – WFP Tanzania | *Human-Centered Design, Figma* Aug 2024 – Jan 2025

- Built and prototyped a collaborative food cargo tracking platform to improve logistics transparency across **10+ delivery points** in Tanzania, enhancing accountability among drivers and vendors and directly supporting **UN SDG 2: Zero Hunger**.
- Designed a **real-time dashboard** integrating **geofencing**, **GPS tracking**, and **AI-powered metrics analysis**, projected to reduce communication inefficiencies and emergency calls by **80%**, enabling faster delivery of food aid to vulnerable communities.

Tiny House Project | *Engineering Project, Python, Excel* Sep 2025 – Present

- Evaluated and optimized battery energy-storage configurations for an off-grid tiny house, analyzing capacity, efficiency, load requirements, and **cost estimates** in **Python** and **Excel** to support reliable power delivery.